# A torsion Z/22 curve over a quartic field

Problèmes ouverts — A. Dujella · 2026

**Problem 2.9 (Dujella, 2026).** Is there an elliptic curve over a quartic field with torsion group $\mathbb{Z}/22\mathbb{Z}$ and positive rank ?

*Remark.* Solved by Maksym Voznyy on 2024/01/04. There is a curve with torsion group $\mathbb{Z}/22\mathbb{Z}$ and rank 2 over the quartic field defined by the polynomial $f(x) = x^4 + \frac{50}{27}x^3 + \frac{281}{243}x^2 + \frac{68}{243}x + \frac{2}{81}$. The curve has the equation

$$y^2-(201204z^3+328374z^2+159732z+8019)xy-(14996874942306z^3+23680026731268z^2+10746821159640z+93126$$
$$= x^3 - (1435244994z^3 + 2266248132z^2 + 1028502360z + 89124624)x^2,$$

where $z$ is a root of $f$.